

Auteur: Ilyas Sefiani
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$$\int_{-1}^0 \frac{dx}{\sqrt{1-x^2}}$$
$$= \lim_{a \rightarrow -1+} \int_a^0 \frac{dx}{\sqrt{1-x^2}}$$

Primitieve vinden:

$$\int \frac{dx}{\sqrt{1-x^2}} = \text{Bgsin}(x)$$
$$= \lim_{a \rightarrow -1+} (\text{Bgsin}(0) - \text{Bgsin}(a))$$
$$= \left(0 + \frac{\pi}{2}\right) = \frac{\pi}{2}$$

Integraal is convergent!

References